# Electromagnetic induction

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain why changing magnetic flux can induce an emf.

**SUCCESS CRITERIA - I can...**

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A **3 Ω** resistor carries **5 A**. **Find power** using both IV and $I^2R$, showing the implied voltage.

**Marzano check - circle:** ① ② ③

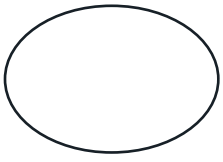
## 02 Worked example: explain the prediction

**Reference:** $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

**Worked reasoning:** A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

**Your turn:** Draw a magnet approaching a coil and **mark the changing field through** the loop. Contrast it with a stationary case.

Label field direction, loop area and what changes.



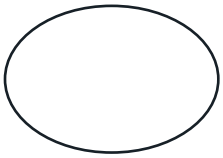
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

1. **Compare a magnet held still** beside a fixed coil with a magnet moving into the coil.
2. Which can produce a changing flux?

Label field direction, loop area and what changes.



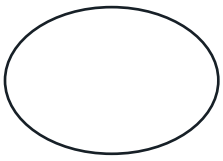
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A **100**-turn lab coil has area **0.020 m²**. A uniform field perpendicular to it changes from **0.10 T** to **0.30 T** in **0.50 s**. **Label the magnetic-flux diagram** and **calculate average induced-emf magnitude**.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

**Marzano check - circle:** 1 2 3

### 06 Review: connect an earlier idea

Both charge magnitudes become **4** times as large and separation becomes **6** times as large. Express new force as a fraction of the old.
**Label charges** and both interaction forces.



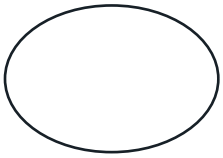

Charge A Charge B

**Marzano check - circle:** 1 2 3

### 07 Exit check

The same flux change occurs in one fourth the time. **Compare emf magnitude** and **name one thing the calculation** treats as true.

Label field direction, loop area and what changes.



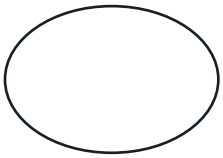
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** 1 2 3

### 08 STRETCH: test the prediction

Flux per turn changes from **0.01** to **0.04 Wb** in **0.2 s** through **50 turns**. **Find average emf magnitude** and **describe Lenz's-law direction qualitatively**.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** 1 2 3